

Anabel Yong

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Summary

MSc in Computational Statistics & Machine Learning from UCL with expertise in Gaussian Processes, Bayesian Optimisation, and kernel-based methods. Interested in machine learning for life sciences discovery and learning biological data.

Publications

- **Yong, A., Tripp, A., Hosseini-Gerami, L., Paige, B. (2025). Bayesian Optimization for Molecules should be Pareto-Aware. (Submitted to NeurIPS 2025 AI for Science Workshop.) Link: [arXiv](#)**

Education

MSc Comp. Statistics & Machine Learning, UCL (Distinction, Dean's List) Sept 2023 - Sept 2024
Modules: Probabilistic & Unsupervised Learning (Gatsby PhD module) (91%), Machine Learning Seminar (Marc Deisenroth- DeepMind)(76%), Numerical Optimization (80%), Statistical Natural Language Processing

BSc Maths & Biology, University of Edinburgh (First Class Honours) October 2019 - June 2023
Modules: Bioinformatics (95%), Machine Learning (80%), Numerical Methods for Differential Equations (85%), Linear Programming, Modelling & Solutions (76%), Mathematics for Natural Sciences (80%)

Work Experience

- **Research Intern, National University of Singapore (PI: Jonathan Scarlett)** May 2024 - Present
 - Implemented biological sequence kernels, Inverse Quadratic Hamming kernels for amino acid sequences.
 - Designing an active learning algorithm on mutation data.
- **Machine Learning Research Intern, Ignotalabs.AI (PIs: Austin Tripp, Cambridge MLG Group & Brooks Paige, UCL)** June 2024 – Sept 2024
 - Developed a novel multi-objective Bayesian optimization algorithm using Tanimoto/MinMax Kernels for small molecule optimization.
- **Research Assistant, Kustatscher Lab, University of Edinburgh** Sept 2022 - July 2023
 - Benchmarked a data pre-processing pipeline for proteomics, pgFDR), to enhance protein detection accuracy.
 - Aided in database curation of human proteins and microproteins of ProteomeHD2
- **Research Intern, ZiHeng Yang Lab, UCL** March 2022 - May 2022
 - Developed Bayesian MCMC algorithms to model ABO blood type frequencies, contributing to population genetics and personalised medicine approaches.
- **Research Intern, Grima Lab, University of Edinburgh** June 2020 - Jan 2022
 - Implemented stochastic and deterministic models to predict mRNA decay kinetics, aiding in the understanding of gene regulation mechanisms and supporting drug efficacy studies in preclinical research.

Skills

Programming: Python, C++, R, MATLAB, git, Linux, Google Cloud Platform (GCP), AWServices (AWS)

Machine Learning: PyTorch, JAX, BoTorch, GPyTorch

Bioinformatics/Cheminformatics: RDKit, PyMOL, UniProt, MaxQuant

References (available upon request)

- **Brooks Paige, Assistant Professor in Machine Learning, University College London:** b.paige@ucl.ac.uk
- **Layla Hosseini-Gerami, PhD Cantab, Chief Data Science Officer, Ignotalabs.ai:** layla.gerami@ignotalabs.ai
- **Austin Tripp, Senior Machine Learning Research Scientist, Valence Labs:** austin.james.tripp@gmail.com